

MELEK DERMAN

Curriculum Vitae

Nuclear Science and Engineering
Oregon State University

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Research Interests

Radiation Transport; Charged-Particle Transport; Deterministic Transport Methods;
Monte Carlo Methods; High-Performance Computing (HPC);
Parallel Computing & Performance Optimization; Computational Physics;
Scientific Software Development; Quantum Computing Applications in Nuclear Engineering

Education

- Sept 2024 – **Ph.D., Nuclear Engineering**, *Oregon State University*, Corvallis, OR.
Present
 - Center for Advancing the Radiation Resilience of Electronics (CARRE), DOE PSAAP IV Project (Grant No. DE-NA0004268)
 - Center for Exascale Monte Carlo Neutron Transport (CEMeNT), DOE PSAAP III Project (Grant No. DE-NA003967)
 - Radiation Transport and Reactor Physics Research Group

May 2024 **MS, Nuclear Engineering**, *Texas A&M University*, College Station, TX.
 - Center for Nuclear Security Science and Policy Initiatives (NSSPI)

June 2013 **BS, Physics with Honors**, *Akdeniz University*, Antalya, Turkey.
 - Nuclear Research and Application Center (NUBA)

Additional Education

- June 2021 **AA, Justice**, *Anadolu University*, Eskisehir, Turkey.
June 2015 **BA, Business Administration**, *Anadolu University*, Eskisehir, Turkey.

Research Experience

Oregon State University, School of Nuclear Science and Engineering, Corvallis, OR

- Sept 2025 – **Graduate Research Assistant**.
Present
 - Implementing the single-scattering electron transport capability in MC/DC to enable the code's first charged-particle transport functionality
 - Designing and implementing the verification and validation (V&V) suite to support the electron transport module

June 2025 – **Graduate Project Assistant**.
Sept 2025
 - Enabled CAD-based geometry input in MC/DC by integrating MC/DC support into GEOUNED, an external CAD converter for Monte Carlo codes
 - Developed and now maintain PyEEDL, a Python tool for converting ENDF-6-formatted EEDL datasets into MC/DC-compatible libraries (<https://doi.org/10.5281/zenodo.18425379>)
 - Contributed to the research, implementation, and validation of the single-scattering electron transport capability

Sept 2024 – **Graduate Research Assistant.**

- June 2025
- Implemented and verified analytical and literature benchmark problems in the MC/DC code, including asymmetric supercritical transients and multigroup reactor-physics cases
 - Configured and deployed AMD's Omnitrace profiling tool on CEMeNT's AMD CPU/GPU systems to support performance analysis and optimization of MC/DC workloads
 - Executed and profiled large-scale MC/DC simulations on HPC clusters, developing automated challenge-problem workflows and performance profiling pipelines
 - Developed and analyzed numerical methods for deterministic transport, including a novel, Dynamic Mode Decomposition-based spectral analysis approach and a PyMFEM/MFEM-based Boltzmann transport solver* using iterative methods (GMRES, Block-Jacobi) on parallel architectures (*<https://doi.org/10.5281/zenodo.15107101>)

Texas A&M University, Department of Nuclear Engineering, College Station, TX

Jan 2023 – **MS Thesis - Nuclear Safeguards Feasibility Study for a Theoretical Molten Salt Reactor (MSR).**
May 2024

- Developed an MCNP model for the MSR designed at Texas A&M University
- Established a comprehensive neutronics model by analyzing the buildup of actinides and fission products
- Designed and developed a nuclear safeguards monitoring approach to ensure the peaceful use of special nuclear material (SNM)

Selected Graduate Level Coursework

- 2025 **Parallel Programming**, *Oregon State University.*
- 2025 **Software Development for Research**, *Oregon State University.*
- 2024 **Introduction to Data Science for Engineers**, *Oregon State University.*
- 2024 **Machine Learning**, *Oregon State University.*
- 2023 **Monte Carlo Methods for Particle Transport**, *Texas A&M University.*
- 2022 **Directed Study (Neutron Transport Theory and Computer Applications in Nuclear Engineering)**, *Texas A&M University.*

Selected Academic & Research Projects

- Feb 2025 **Team Member - High-Performance Neutron Transport on Heterogeneous Architectures (AMD Tools Workshop, Eugene, OR).**
Optimized the MC/DC neutron transport code on AMD MI300A APUs using Python and Numba; integrated AMD profiling tools to identify execution hotspots in LLVM-IR compiled kernels; and worked on porting Harmonize runtime's asynchronous scheduling logic to AMD hardware to minimize thread divergence
- Nov 2023 **Team Leader - Multi-Purpose Canister (MPC) Modelling and Simulations (Texas A&M University, College Station, TX).**
Developed MCNP models of an MPC; and performed Columbia River full-submersion radiation transport simulations to evaluate dose fields, shielding behavior, and potential radionuclide release
- May 2023 **Team Leader - Fast Breeder Reactor (FBR) Core Modeling & Safeguards Analysis (Texas A&M University, College Station, TX).**
Led MCNP-based neutronic modeling of an FBR core; and developed a preliminary safeguards assessment to evaluate diversion pathways and SNM monitoring signatures

Technical Skills

Nuclear Engineering Software: MCNP6, Visual Editor for MCNP6, OpenMC, SCALE6 (basic), INCC, GENIE, InterSpec

Programming Languages: Python (NumPy, SciPy, pandas, Qiskit, scikit-learn, PyMFEM, Numba), C++ (basic), MATLAB, Mathematica

Scientific Computing & HPC: Linux HPC systems; Bash; parallel computing (OpenMP, CUDA, OpenCL, MPI/mpi4py); job scheduling (Slurm, Flux)

Other Tools: L^AT_EX, Git/GitHub, Microsoft Office, Visio, Adobe Photoshop, Adobe Illustrator, Windows/Linux/macOS

Publications

- 2026 W. Bennett, R. G. McClarren, E. Smith, **M. Derman**, Weak-DMD: A Galerkin approach to the problem of noise in the Dynamic Mode Decomposition algorithm, Preprint. [dx.doi.org/10.2139/ssrn.6515172](https://doi.org/10.2139/ssrn.6515172).
- 2026 **M. Derman**, J. P. Morgan, K. E. Niemeyer, T. S. Palmer, Dynamic Spectral Analysis of Time-Dependent One-Cell Inversion Transport Method Using Windowed DMD, **Manuscript to be submitted to *Nuclear Science and Engineering***.
- 2013 I. Boztosun, H. Djapo, S.F. Ozmen, Y. Cecen, M. Karakoc, A. Coban, A. Cesur, T. Caner, E. Bayram, G.B. Keller, B. Kucuk, A. Guvendi, **M. Derman**, D. Kaya, The Results of the First Photoneuclear Reaction Performed in Turkey: The Zinc Example, In *Turkish Journal of Physics*.

Scientific Software

- 2026 **M. Derman**, PyEPICS: Python library for reading, converting, and querying the LLNL EPICS nuclear data, Zenodo. doi.org/10.5281/zenodo.18648203.
- 2025 **M. Derman**, PyEEDL: Python tool for converting ENDF-6 formatted EEDL datasets into MC/DC-compatible datasets, Zenodo. doi.org/10.5281/zenodo.18425379.

Scholarships, & Awards

- Sept 2024 **Provost's Distinguished Graduate Scholarship**, Oregon State University
- Jan 2022 **Graduate Scholarship**, Ministry of National Education of the Republic of Turkey
- Mar 2013 **1st Place Award** - Ion Optics and Gamma-Ray Detection Experiment, Heavy Ion Laboratory, University of Warsaw, Poland

National Laboratory Trainings

- Apr 2024 **Office of International Nuclear Safeguards Nonproliferation Workshop**, Oak Ridge National Laboratory, Oak Ridge, TN.
- Mar 2024 **SEE LANL Nondestructive Assay Training Course**, Los Alamos National Laboratory, NM.

Memberships

- 2023 - Present **Student Member**, American Nuclear Society (ANS).
- 2024 - Present **Student Member**, World Institute for Nuclear Security (WINS).
- 2023 - 2024 **Student Member**, Institute of Nuclear Materials Management (INMM).
- 2022 - 2024 **Student Member**, Nuclear Security Science & Policy Institute (NSSPI) at Texas A&M University.

Other Professional Experience

Turkish Statistical Institute, Ankara, Turkey

June 2021 – **HSE Specialist.**

Sept 2021 Advised management on Occupational Health and Safety (OHS) regulations in response to Covid-19; prepared risk assessment reports; and delivered safety training programs

Aug 2016 – **HR Specialist.**

Jan 2022 Assisted in managing the \$50M+ annual budget operations by monitoring expenses, preparing financial tables, and reporting variances

Muratpasa Anatolian High School, Antalya, Turkey

Jan 2015 – **Teaching Intern.**

Nov 2015 Instructed high school physics courses in adherence to the curriculum; administered exams and prepared progress reports; and conducted parent-teacher meetings to discuss academic performance

Freelance, Antalya, Turkey

May 2008 – **Private Tutor.**

Jan 2015 Delivered private tutoring in Physics and Mathematics to students ranging from primary to high school; prepared customized lesson plans; supported exam preparation; and reinforced fundamental problem-solving skills for competitive exams

Akdeniz University, Department of Health, Culture and Sports, Antalya, Turkey

Jan 2008 – **Graphic Design Assistant (Part-Time).**

Nov 2008 Designed promotional materials (banners, posters, brochures) for university events using Adobe Photoshop and CorelDRAW.

Other Certifications

May 2017 **Occupational Health and Safety Specialist Certification**, The Turkish Ministry of Labor and Social Security.

Jan 2016 **Pedagogical Formation Certificate in Physics Teaching**, *One-year teacher education and practicum program for qualification as a high-school teacher*, School of Education, Akdeniz University, Antalya, Turkey.